Sri Lanka Institute of Information Technology

Research Project- IT4100

PROJECT CHARTER

2026, Semester 1

GROUP ID: R26-IT-086



1. Project overview

SpeakFree is a four-component SLIIT final year research project that builds an AI powered online platform for affordable speech disorder screening and privacy preserving therapy support.

Project title	SpeakFree: AI Online Speech Disorder Screening and Therapy Assistant
Project ID	R26-IT-086
Institution	Sri Lanka Institute of Information Technology (SLIIT)
Department	Department of Information Technology
Module	Research Project, 2026 Regular Batch
Project window	March 2026 (proposal) to December 2026 (publication evidence)
Supervisor	Ms. Chathurangika Kahandawaarachchi
Co-supervisor	Ms. Suranjini Silva
Team size	4 members, one component each
Charter version	1.0, draft, 15 October 2026

2. Background and problem statement

Many people cannot reach a qualified speech language therapist, and the automatic speech recognition (ASR) tools that could help perform poorly on disordered speech. The problem has five connected parts.

Problem	What it causes	How Speak Free responds
Manual screening is slow, difficult and expensive	Long waits, scoring that depends on human judgment, and a need for trained staff and expert time	Automated online screening that returns scores within one session
Access is limited for rural and remote communities	Long travel to a clinic, few therapists, long waitlists and unstable connections interrupt care	A browser based assessment that works from home, with practice support between visits
Early symptoms are missed in routine sessions	Subtle signs such as sound substitutions (w for r), cluster simplification, vowel distortion, weak syllable stress and irregular intonation go unnoticed	Acoustic, phoneme level and prosody analysis that surfaces what the ear can miss

Problem	What it causes	How Speak Free responds
Basic practice apps stop at pronunciation drills	No AI scoring, articulation analysis, severity prediction or therapy recommendation	A complete screening to therapy pipeline across four components
Results are unstructured and inconsistent	Scattered notes and low confidence make decisions and progress tracking hard	Structured, explainable results with scores, severity, findings and recommendations

The research literature adds three further gaps:

- Standard ASR systems show error rates reported as high as 60 to 80 percent on disordered speech.
- Existing screening tools are tested mostly in clean lab conditions, with little evaluation in noisy, real world settings.
- Single-task elicitation, such as read aloud passages, routinely underestimates severity in speakers whose speech varies between tasks. SpeakFree therefore uses three assessment tasks: audio recording of a passage, picture description and scenario description.
- Speech recordings are sensitive personal data, so privacy must be protected at every stage.

3. Purpose and justification

The project exists to make speech disorder screening and therapy follow-up affordable and reachable online, without giving up the privacy of the people being screened.

- **For people who need screening:** a low-cost first assessment from a browser, without a clinic visit.
- **For therapists:** structured fluency and articulation evidence, plus privacy preserving cohort reports, to support clinical decisions.
- **For research:** a screening to therapy pipeline evaluated on noisy, real world style speech rather than clean lab recordings only.
- **For the module:** a working, integrated research product and an academic record of its methodology, evaluation and findings.

4. Objectives and success criteria

The main objective is to deliver an integrated, privacy preserving pipeline that screens speech, analyses articulation, classifies severity and supports home therapy practice.

Component	Objective	Success criterion
1. Speech screening and feature extraction	Capture audio in the browser, reduce noise, and produce a structured fluency profile from confidence aware ASR (Whisper), HuBERT embeddings and Librosa acoustic features	A working pipeline that outputs a privacy respecting fluency profile and hands clean features to Component 2
2. Articulation and alignment analysis	Detect articulation errors through forced alignment and phoneme level analysis of the features from Component 1	Phoneme level error indicators produced for screened recordings and passed to Component 3
3. Severity classification	Combine fluency and articulation signals into one overall severity class	A classifier that returns a severity class from Component 1 and 2 outputs, with its accuracy reported
4. AI therapy assistant and adaptive learning	Support home practice with an expert recalibratable scoring model, a contextual bandit exercise planner, a conversational assistant and differential privacy cohort reports	Scoring validated against expert judgment, adaptive plans generated, and therapist cohort reports produced with privacy protection

Project level success criteria:

1. All four components built and integrated end to end before the final presentation.
2. Every milestone in section 9 submitted on or before its date.
3. A research paper and final thesis submitted, with publication evidence provided by 1 December 2026.
4. No raw speech recording exposed outside the privacy controls each component defines.

5. Scope

The project covers screening, analysis, severity classification and therapy support as a research prototype. It does not replace a clinician.

In scope

- Browser based audio capture, noise reduction and acoustic feature extraction
- Confidence aware ASR and self-supervised speech embeddings
- Forced alignment, phoneme level segmentation and articulation error detection
- Severity classification from fluency and articulation signals
- Adaptive home practice planning, an assistant, and differential privacy cohort reporting
- Evaluation on noisy, real world style speech, and expert validation of therapy scoring
- An academic project website, thesis, research paper and presentations

Out of scope

- Clinical diagnosis, or any claim that the platform replaces a speech language therapist
- Regulatory approval or clinical trials
- Production hosting, commercial support and long-term maintenance beyond the module
- Languages and speech disorders not covered by the chosen datasets

6. Solution overview and architecture

SpeakFree is a three-tier web application. A React frontend collects speech through guided tasks, a FastAPI backend handles uploads, authentication and task management, and an AI analysis pipeline returns scores, severity and therapy recommendations that are stored in a PostgreSQL database.

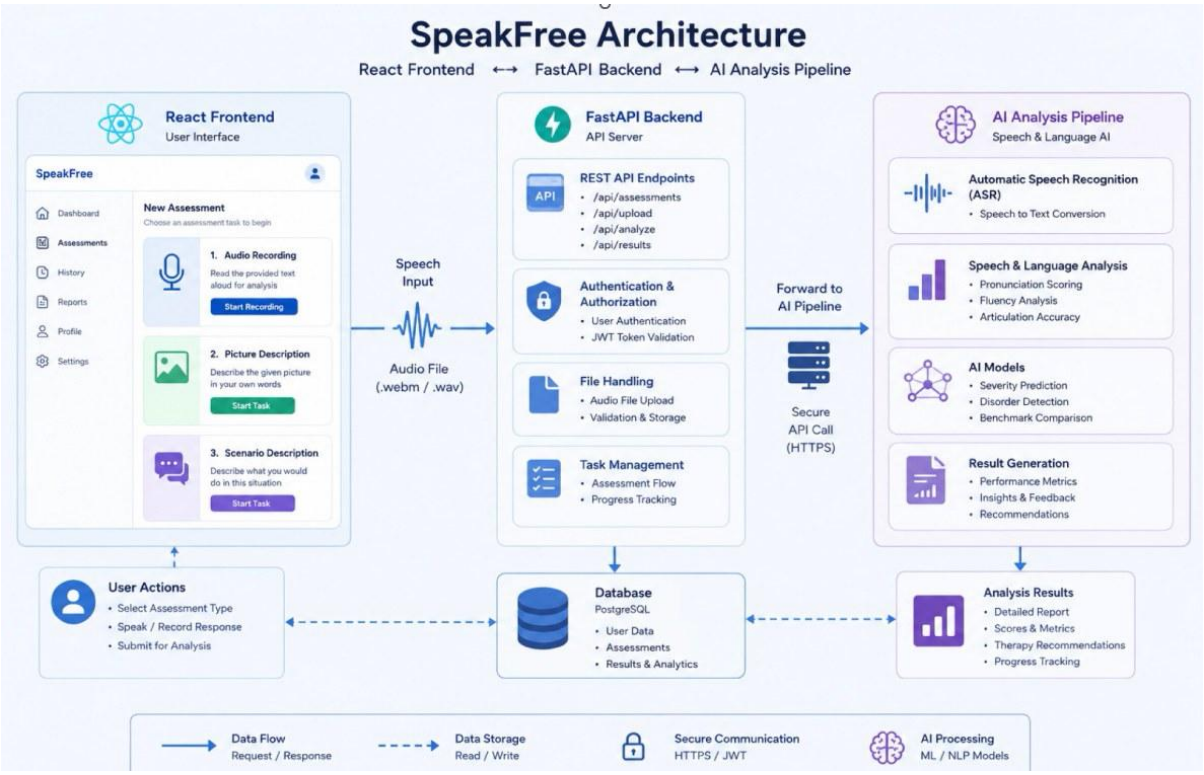


Figure 1. SpeakFree architecture: React frontend, FastAPI backend and AI analysis pipeline (concept diagram).

How a session flows:

1. The user selects an assessment type and records or describes a response in the browser (audio saved as .webm or .wav).
2. The frontend sends the audio to the backend over HTTPS. The backend validates and stores the file, and checks the user's JWT token.
3. The backend forwards the audio to the AI pipeline: ASR, speech and language analysis, severity and disorder models, and result generation.
4. Results (scores, findings, recommendations, progress) are saved to the database and shown to the user and, where permitted, the therapist.

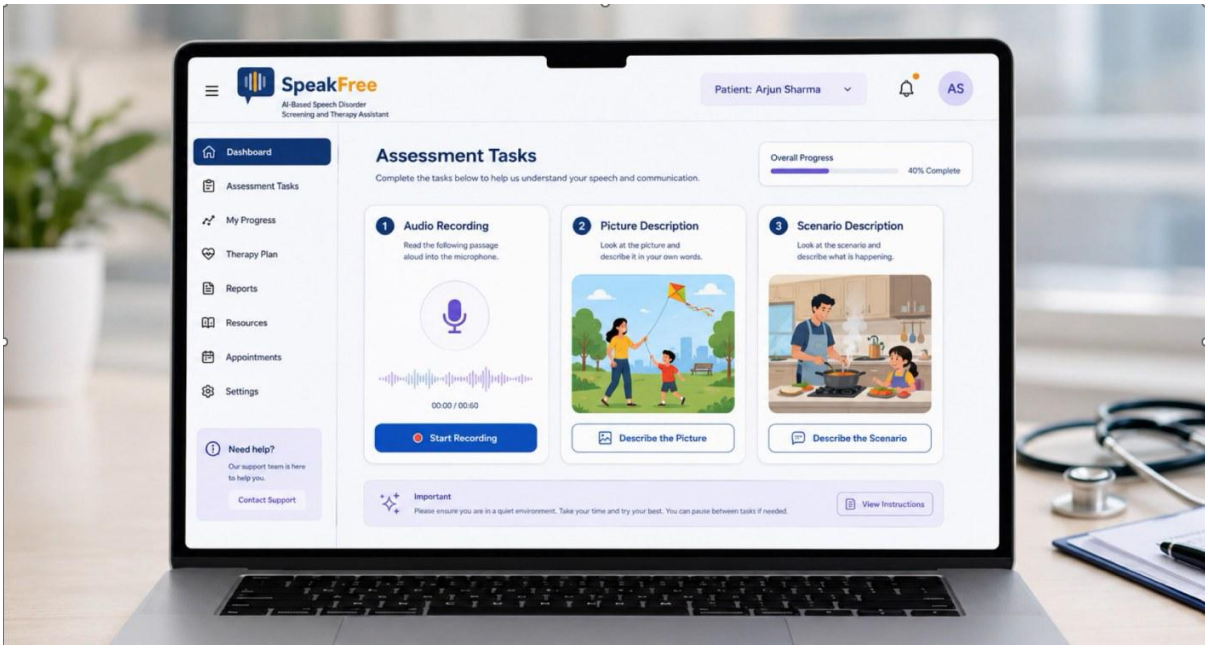


Figure 2. Assessment tasks screen: audio recording, picture description and scenario description (concept mock-up).

7. Components, team and responsibilities

Each of the four members owns one component and is accountable for its design, build, evaluation and individual thesis.

Component	Owner	Student ID	Responsibility
1. Speech screening and feature extraction	P.A.S.S. Kusumsiri	IT22132628	Audio capture, noise reduction, ASR, embeddings, acoustic features, fluency profile
2. Articulation and alignment analysis	E.Y. Randiw	IT22257086	Forced alignment, phoneme segmentation and articulation error detection
3. Severity classification	I.L.S. Tharushan	IT21315114	Severity model combining fluency and articulation signals
4. AI therapy assistant and adaptive learning	M.D.I. Perera	IT22056702	Scoring model, exercise planner, assistant, privacy preserving cohort reports
Project supervision	Ms. Chathurangika Kahandawaarachchi	n/a	Guidance, milestone review, assessment of deliverables

Shared team responsibilities:

- Agree the interfaces between components: features handed from 1 to 2, signals from 1 and 2 to 3, and outcomes to 4.
- Keep a common repository, logbook and milestone tracker current.
- Prepare the common report, website, presentations and demonstration together.

Component concept visuals

The figures below are concept designs for each component's output. Numbers shown in them are illustrative placeholders, not measured results.

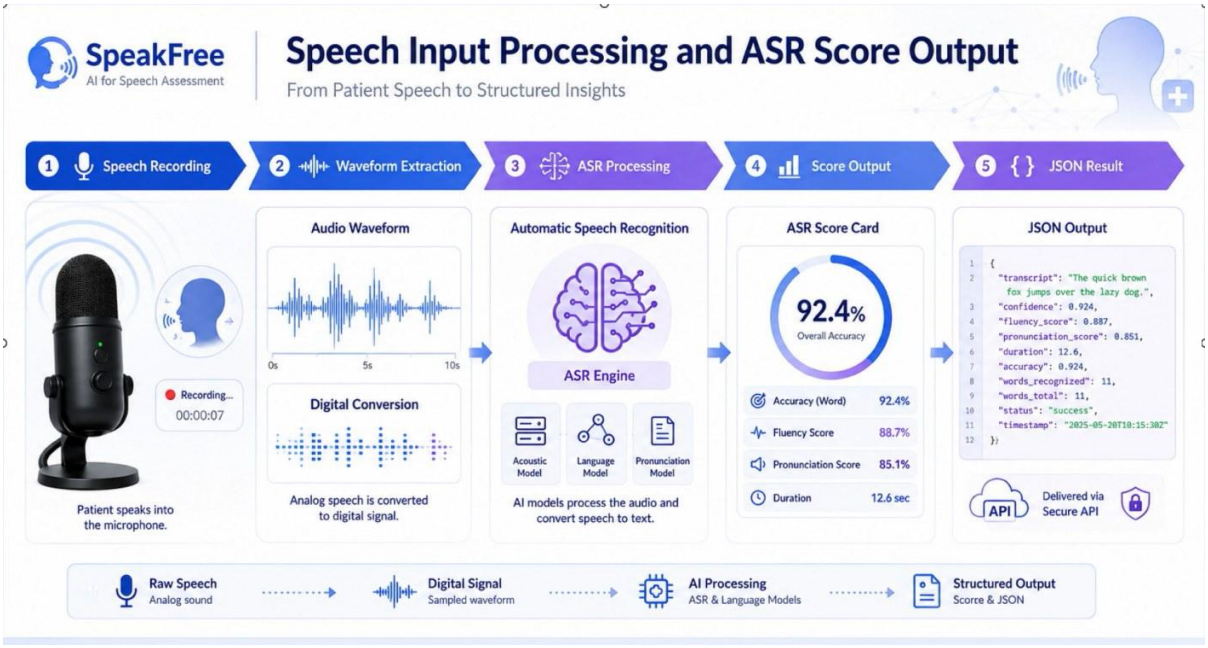


Figure 3. Component 1: speech input processing and ASR score output.

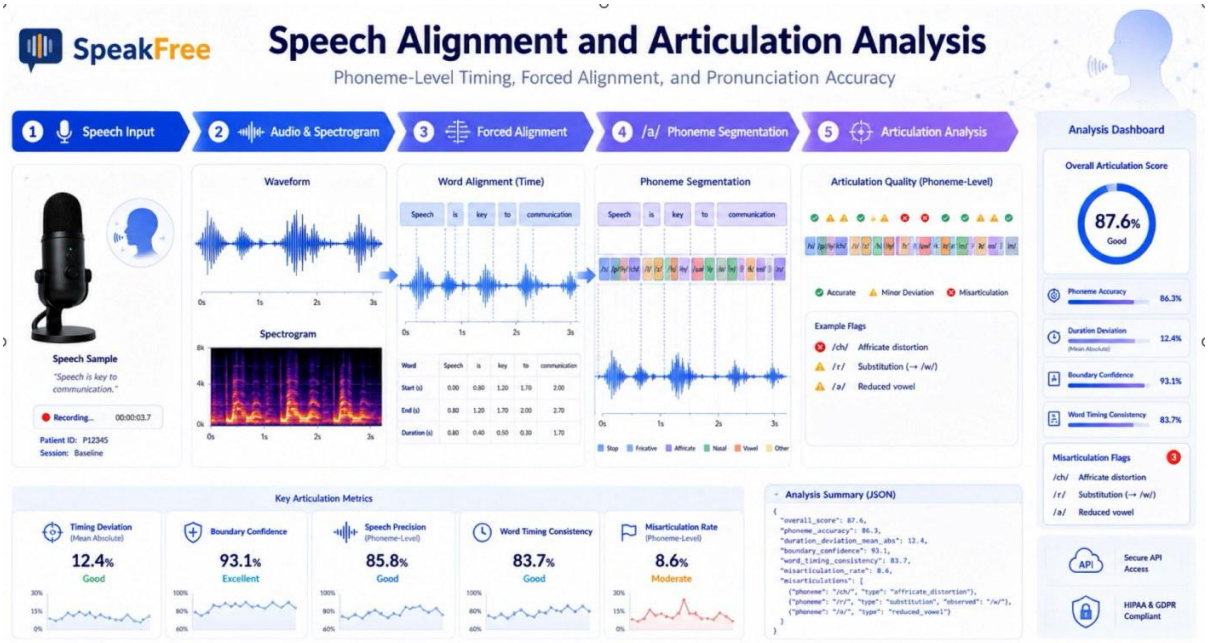


Figure 4. Component 2: speech alignment and articulation analysis.

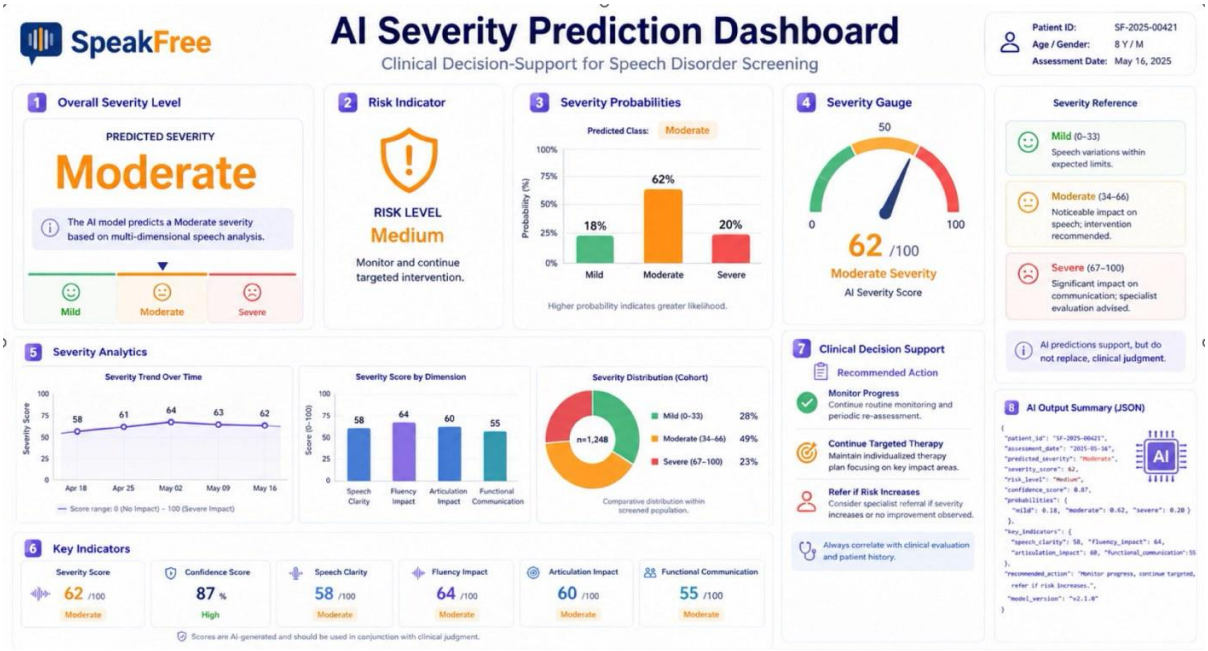


Figure 5. Component 3: AI severity prediction dashboard.



Figure 6. Component 4: personalised therapy recommendations and progress tracking.

8. Deliverables

The project hands over working software, written records and presentations, each tied to a milestone in section 9.

Deliverable	Description	Due
Project proposal	Problem, objectives, methodology and plan	15 March 2026 (submitted)
Draft thesis	Individual thesis draft for each component	11 October 2026
Project website	Academic information site: Home, Domain, Milestones, Documents, Presentations, About Us, Contact Us	11 October 2026
Working platform	Four integrated components with demonstration	19 to 21 October 2026
Logbook	Record of weekly project activity	19 to 21 October 2026
Research paper	Conference style paper on the project	23 October 2026
Final thesis	Final individual thesis and common report	28 October 2026
Publication evidence	Proof of paper submission or publication	1 December 2026

9. Milestones and timeline

Dates follow the 2026 Regular Batch research project timeline. The ICAC conference paper deadline of 1 September 2026 is a separate, earlier conference deadline from the module's own paper submission on 23 October 2026.

#	Milestone	Date (2026)	Notes
1	Proposal submission	15 March	
2	Proposal presentation	16 to 18 March	
3	Progress presentation 1	11 to 13 May	
4	Checklist submission	13 May	
5	Progress presentation 2	31 August to 2 September	
6	Draft thesis submission	11 October	
7	Website submission	11 October	
8	Final checklist submission	14 October	

#	Milestone	Date (2026)	Notes
9	Final presentation and viva	19 to 21 October	Demonstration 10 min, viva 35 min, marking 5 min
10	Website evaluation and logbook submission	19 to 21 October	
11	Research paper submission	23 October	
12	Final thesis submission	28 October	
13	Research paper publication evidence submission	1 December	

Critical path. The integrated demonstration (19 to 21 October) depends on all four components handing results to each other, and the final thesis (28 October) depends on the viva feedback. Any slip in a component's interface delays the whole pipeline.

10. Technology and resources

Layer	Technology
Frontend	React web application with guided assessment tasks, dashboards, progress and reports
Backend	FastAPI server with REST endpoints (/api/assessments, /api/upload, /api/analyze, /api/results), JWT authentication, audio file handling and task management
Database	PostgreSQL for user data, assessments, results and analytics
Speech and AI	Whisper (confidence aware ASR), HuBERT embeddings, Librosa acoustic features, forced alignment, severity and disorder models, contextual bandit planner, differential privacy reporting
Security	HTTPS for all calls, JWT token validation, privacy preserving feature handoff between components
Tooling	VS Code, Git and GitHub for source control, GitHub Pages for the project website

Other resources: supervisor guidance, speech datasets and expert (therapist) review for validation, and a shared project repository. Funding.

11. Assumptions and constraints

Assumptions

- Suitable speech datasets and expert reviewers are available for training and validation.
- Each member completes their component on time and agrees the interfaces early.
- Participants use a quiet environment and a working microphone and browser.
- Open source libraries and models are sufficient, with no paid services required.

Constraints

- Fixed academic timeline: final thesis due 28 October 2026 and publication evidence due 1 December 2026.
- The project website must stay within the module's limit of 20 MB and use only WordPress, HTML and CSS.
- Speech data is sensitive and must follow the privacy controls in each component.
- Small team of four, with one owner per component and limited testing time near the viva.

12. Risks and mitigation

Risk	Impact	Likelihood	Mitigation
ASR errors on disordered speech	High	High	Confidence aware ASR, HuBERT embeddings, and report error rates openly
Noisy real world recordings degrade results	High	Medium	Noise reduction in Component 1 and testing on noisy samples
Interface mismatch between components	High	Medium	Agree data formats early and test the handoff chain end to end before the viva
Limited labelled or expert data for validation	Medium	Medium	Early expert review, recalibratable scoring and clear limits in the thesis
Privacy breach of speech recordings	High	Low	HTTPS, JWT, minimal storage, differential privacy for cohort reports
Schedule pressure around 19 to 28 October	High	High	Freeze features early, finish integration before the viva, split writing work
Team member unavailable	Medium	Low	Document each component and share the repository access

13. Stakeholders, governance and communication

Stakeholder	Interest	Involvement
Research team (4 members)	Deliver a working, assessed project	Build, test, report and present
Supervisor	Quality and progress of the research	Regular guidance and milestone review
Examiners and viva panel	Assess the project against module criteria	Presentations, viva, thesis and website evaluation
Speech language therapists	Clinically useful, explainable results	Expert validation and feedback
Patients and families	Affordable, easy and private screening	Prospective users of the platform
SLIIT	Academic standards and research output	Module oversight

Governance and communication

- Scope, interface and schedule decisions are made by the team and confirmed with the supervisor; progress is tracked in the logbook and the milestone table in section 9.
- The team meets weekly and meets the supervisor once a month.
- Any change to scope or a milestone date is recorded in a change note and agreed with the supervisor before it takes effect.